1. Topic: Structure of the Earth

Scenario

In 2025, a team of geologists discovered unusual seismic activity in the Albertine Rift Valley, Uganda. Data revealed that the Earth's crust in the region is thinning at an unprecedented rate of 2 cm/year, accompanied by frequent tremors. Satellite imagery shows new fissures emitting geothermal gases, and local communities report dried-up water tables. Scientists hypothesize that this could lead to the formation of a new ocean basin over the next 50 million years.

Task

Critically evaluate the potential socio-economic and environmental impacts of this geological phenomenon on Uganda. Propose a sustainable land-use plan for

the affected communities, integrating knowledge of the Earth's internal structure and tectonic processes. Justify your plan with reference to Uganda's development goals.

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2. Topic: Continental Drift Theory
Scenario:
Recent paleomagnetic studies in Antarctica uncovered coal deposits identical to those in India, suggesting they were once part of the same landmass. Meanwhile, GPS data shows that Australia is drifting northward at 7 cm/year, while the Himalayas are rising by 5 mm/year. A debate has emerged among geographers: *Is the Continental Drift Theory still relevant in explaining current landform changes, or should it be revised?*

Task:
Analyze the evidence provided and argue whether the Continental Drift Theory adequately explains modern geodynamic processes. Recommend how Uganda's education curriculum should teach this theory, considering new technological advancements like GIS.

3. Topic: World Development
Scenario:
The UNDP's 2025 report highlights that Uganda's Human Development Index (HDI) rose to 0.55, but regional disparities persist. While Kampala boasts tech hubs and skyscrapers, Karamoja faces chronic food insecurity. A UN official proposes redirecting 30% of Uganda's oil revenues to equitable regional development, but critics argue this would deter foreign investors.

Task:
Assess the effectiveness of the UN proposal using *three* indicators of development from the syllabus. Design an alternative policy that balances economic growth and equity, referencing Uganda's Vision 2040.

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4. Topic: Agriculture and Development
Scenario:
A Ugandan farmer, Mr. Kato, practices mixed farming but faces declining yields due to soil degradation. The government promotes agroecology, urging farmers to adopt zero-tillage and permaculture. However, Kato argues, "These methods are too slow to feed my family." Meanwhile, commercial farms in Nakasongola use AI-

driven precision agriculture, doubling productivity.

Task:
Evaluate the trade-offs between subsistence agroecology and high-tech commercial farming in Uganda. Recommend a hybrid model for smallholders like Kato, ensuring alignment with SDG 2 (Zero Hunger).

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5. Topic: Soil Degradation
Scenario:
In Mbale District, landslides triggered by deforestation have destroyed 500 homes. Soil tests reveal that the topsoil's pH has dropped to 4.5, rendering it infertile. The government plans to enforce afforestation, but locals resist, citing land shortages for

cultivation.

Task
Propose a multi-stakeholder soil conservation strategy for Mbale, integrating scientific (e.g., terracing) and socio-economic (e.g., land tenure reforms) solutions. Use a flowchart to illustrate your plan.

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6. Topic: Population and Migration
Scenario:
Uganda’s refugee population exceeds 1.5 million, straining resources in Adjumani. Host communities complain of overcrowded schools and depleted firewood, while refugees accuse locals of discrimination. The UNHCR suggests creating “refugee cities” with segregated

services, but human rights groups oppose this as apartheid-like.

Task:
Using push-pull factors and dependency ratios, analyze the long-term viability of the UNHCR's proposal. Draft a manifesto for inclusive integration, citing Uganda's Refugee Act (2006).

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7. Topic: Drainage Systems
Scenario:
The Nile Basin Initiative reports that Uganda's wetlands have shrunk by 40% since 2000 due to urban expansion. A Kampala developer plans to drain Nakivubo Wetland for a mall, arguing it will create jobs. Environmentalists warn this would exacerbate flooding, as seen in 2022 when 10,000 homes were

submerged.

Task:
Conduct a cost-benefit analysis of the mall project, considering hydrology (e.g., water retention capacity) and SDG 11 (Sustainable Cities). Recommend whether the government should approve the project, with mitigations.

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8. Topic: Energy Production
Scenario:
Uganda's oil extraction in Hoima has begun, but leaked reports reveal that TotalEnergies' pipelines could displace 10,000 people. Renewable energy advocates propose geothermal plants in the Rift Valley, but investors cite high upfront costs.

Task:
Compare the feasibility of oil versus geothermal energy for Uganda’s development, using the *energy trilemma* (security, equity, sustainability). Design a 20-year transition plan for the government.

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9. Topic: Climatology
Scenario:
Lake Victoria’s water temperature has risen by 2°C since 2000, causing erratic rainfall. Fishermen report declining tilapia catches, while farmers in Masaka face prolonged droughts. A climatologist attributes this to Indian Ocean Dipole shifts, but policymakers blame local deforestation.

Task:
Distinguish between global and local causes of Lake Victoria's climate changes. Outline an adaptation strategy for fishermen and farmers, using ICT tools like satellite weather data.

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10. Topic: Urban Settlement
Scenario:
Kampala's population is projected to reach 10 million by 2035, but 60% live in informal settlements with no sanitation. A city planner proposes "vertical slums" (high-rises for low-income earners), while NGOs advocate for decentralized "satellite towns."

Task:

Critique both proposals using *Burgess's Concentric Zone Model* and *smart city* principles. Sketch a sustainable urban plan for Kampala, labeling zones for green spaces, industry, and affordable housing.

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11. Topic: Slope Processes
Scenario:
In Kigezi, terraced hillsides are collapsing due to intense rainfall from climate change. Elders recall traditional *ngitiri* (grass strips) stabilized slopes, but youth prefer concrete retaining walls.

Task:
Evaluate the effectiveness of indigenous versus *modern* slope stabilization methods. Justify your preferred approach with geomorphological evidence.

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12. Topic: Food Security
Scenario:
Uganda’s maize exports to South Sudan dropped after aflatoxin contamination was detected. The EU now rejects Ugandan grains, costing $300 million annually. The Ministry of Agriculture blames poor post-harvest handling, but farmers accuse middlemen of storing grain in damp warehouses.

Task:
Devise a value-chain intervention to restore Uganda’s grain trade, linking soil management (Topic 5), storage tech, and policy enforcement.

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13. Topic: Coastal Landforms
Scenario:
Lake Albert's shores are receding by 15 m/ year due to sand mining and wave action. A Chinese firm proposes building a seawall, but ecologists warn it would destroy fish breeding grounds.

Task:
Assess the seawall proposal using costal erosion theories. Propose an eco-friendly alternative, such as artificial reefs, with diagrams.

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14. Topic: Natural Resources
Scenario:
Uganda's rare earth minerals (e.g., niobium) are critical for electric vehicles,

but mining would displace Bwindi's gorillas. A presidential advisor suggests "conservation concessions," where miners pay to protect habitats.

Task:
Debate the ethics of "green mining" in Uganda, referencing *resource curse* theory and SDG 15 (Life on Land).

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15. Topic: Climate Change
Scenario:
Mt. Rwenzori's glaciers have vanished, affecting River Semliki's flow. Hydropower dams now operate at 30% capacity, causing blackouts. A climate activist demands reparations from industrialized nations, but the Energy Minister prioritizes fossil fuels as a "quick fix."

Task:
Analyze Uganda’s energy dilemma using *climate justice* principles. Draft a speech for the Minister, balancing national needs and global commitments.
